A family of counterexamples to the cube-constant conjecture for John ellipsoids of revolution

Counterexample packet for `fa_banach_001`

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Status. Candidate full disproof of Conjecture 5.10 in arXiv:2507.06947, subject to human review. For every pair $1 \le s < d$ for which $s \nmid d$, an explicit centrally symmetric polytope violates the proposed volume bound. The smallest example is $(d, s) = (3, 2)$.

Source conjecture

G. Ivanov, Z. Lángi, M. Naszódi, and À. Sagmeister, *John Ellipsoids of Revolution*, arXiv:2507.06947v2 (5 August 2025) [1]. Conjecture 5.10 on printed and PDF page 19 states that if an origin-centered ellipsoid E solves the fixed-axis problem for a centrally symmetric convex body $K \subset \mathbb{R}^d$ and an s-dimensional subspace F, then

$$\frac{\mathrm{vol}_s(K \cap F)}{\mathrm{vol}_s(E \cap F)} \le C_\square(d, s)\frac{\mathrm{vol}_s([-1,1]^s)}{\mathrm{vol}_s(\mathbb{B}^s)}, \tag{1}$$

where, on writing $d = \ell s + t$, $0 \le t < s$,

$$C_\square(d, s)^2 = (\ell + 1)^t \ell^{s-t}. \tag{2}$$

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Conjecture 5.10. *Let an origin-centered ellipsoid E be a solution to Problem 1 for a centrally symmetric convex body K and a linear subspace F in $\mathbb{R}^d$ of dimension $s \in [d]$. Then*

$$\frac{\mathrm{vol}_s(K \cap F)}{\mathrm{vol}_s(E \cap F)} \le C_\square(d, s)\frac{\mathrm{vol}_s \square^s}{\mathrm{vol}_s \mathbf{B}^s},$$

where the constant $C_\square(d, s)$ is given by

$$C_\square^2(d, s) = \left\lceil \frac{d}{s} \right\rceil^{d - s\lfloor d/s \rfloor} \left\lfloor \frac{d}{s} \right\rfloor^{s - (d - s\lfloor d/s \rfloor)}.$$

The constant $C_\square(d, s)$ in Conjecture 5.10 is attained on certain sections of the cube $\square^d = [-1, 1]^d$ by s-dimensional subspaces (see Lemma 2.2 of [IT21]). Peculiarly, in the case when K is centrally symmetric and s divides d, both our volume ratio bound in F and the inradius bound in $F^\perp$ are simultaneously tight: K is the cube $[-1, 1]^d$; E is the John ellipsoid $\mathbf{B}^d$ of K; F is the solution set to the system of linear equations:

$$\left\langle x, e_{\frac{d}{s}(i-1)+1} \right\rangle = \ldots = \left\langle x, e_{\frac{d}{s}i} \right\rangle, \quad \text{for all} \quad i \in [s].$$

Moreover, the bound $\sqrt{\frac{d}{s}}$ on the inradius in $F^\perp$ is attained on certain sections of the cube

Definitions and normalization

Write $\mathbb{B}^d = \{x \in \mathbb{R}^d : \|x\|_2 \le 1\}$. In the source, an F-operator preserves both F and $F^\perp$, acts as a nonzero scalar on $F^\perp$, and is arbitrary and invertible on F. Thus the identity is an F-operator, and $\mathbb{B}^d$ is an admissible ellipsoid of revolution for every F. We prove below that it is largest even among *all* ellipsoids in the constructed body, which is stronger than the hypothesis required in (1).

For vectors $u, v \in \mathbb{R}^d$, $u \otimes v$ denotes the rank-one operator $x \mapsto \langle x, v\rangle u$.

Proof intuition

The conjectured constant contains an integer partition of d: its square is the largest product of s positive integer block sizes summing to d. John's contact decomposition has positive *real* weights and does not impose this integrality. We divide the total contact weight equally among the s coordinate directions of F, assigning the real weight d/s to each. Suitable lifts into $F^\perp$ make the lifted unit normals an exact John decomposition of the identity. The resulting slab polytope has section $\sqrt{d/s}\,[-1,1]^s$, so it attains the larger continuous Brascamp–Lieb bound from the source. When $s \nmid d$, strict AM–GM says that this continuous value exceeds the integer cube constant.

Counterexample theorem

Theorem 1. *Let $d \ge 3$ and $1 \le s < d$, and assume $s \nmid d$. There are an s-dimensional subspace $F \subset \mathbb{R}^d$ and a centrally symmetric convex polytope $K \subset \mathbb{R}^d$ such that:*

1. *$\mathbb{B}^d \subset K$, and $\mathbb{B}^d$ is a largest-volume ellipsoid in K; in particular $E = \mathbb{B}^d$ solves the source's fixed-axis ellipsoid-of-revolution problem for K and F;*
2. *$K \cap F = \sqrt{d/s}\,[-1,1]^s$ after identifying F with $\mathbb{R}^s$;*
3. $$\frac{\mathrm{vol}_s(K \cap F)}{\mathrm{vol}_s(E \cap F)} > C_\square(d,s)\frac{\mathrm{vol}_s([-1,1]^s)}{\mathrm{vol}_s(\mathbb{B}^s)}.$$

Consequently Conjecture 5.10 is false for every nondivisible pair $1 \le s < d$.

Proof. Put $r = d - s$, and choose orthonormal bases $\{e_1, \dots, e_s\}$ of F and $\{f_1, \dots, f_r\}$ of $F^\perp$. Set

$$p = \sqrt{\frac{s}{d}}, \qquad q = \sqrt{\frac{r}{d}}.$$

For $1 \le j \le s$, $1 \le k \le r$, and $\varepsilon, \delta \in \{-1, 1\}$, define

$$u_{jk}^{\varepsilon,\delta} = \varepsilon p e_j + \delta q f_k.$$

Every such vector is a unit vector. Let

$$K = \bigcap_{j,k,\varepsilon,\delta} \left\{ x \in \mathbb{R}^d : \left|\langle x, u_{jk}^{\varepsilon,\delta}\rangle\right| \le 1 \right\}. \tag{3}$$

This is centrally symmetric. The normals span $\mathbb{R}^d$, since sums and differences of the two choices of δ yield multiples of e_j and f_k. Hence K is bounded and is a convex body. Since every normal is unit, Cauchy–Schwarz gives $\mathbb{B}^d \subset K$.

We next prove the maximal-ellipsoid assertion directly. Give every indexed normal the weight

$$\alpha = \frac{d}{4sr}.$$

The sum of the weights is d, and the sign symmetry makes the weighted vector sum zero. For fixed j, k, cancellation of the cross terms gives

$$\sum_{\varepsilon,\delta} u_{jk}^{\varepsilon,\delta} \otimes u_{jk}^{\varepsilon,\delta} = 4p^2 e_j \otimes e_j + 4q^2 f_k \otimes f_k.$$

Therefore

$$\begin{aligned}\sum_{j,k,\varepsilon,\delta} \alpha\, u_{jk}^{\varepsilon,\delta} \otimes u_{jk}^{\varepsilon,\delta} &= \frac{dp^2}{s} \sum_{j=1}^{s} e_j \otimes e_j + \frac{dq^2}{r} \sum_{k=1}^{r} f_k \otimes f_k \\ &= \mathrm{Id}_F + \mathrm{Id}_{F^\perp} = \mathrm{Id}_{\mathbb{R}^d}.\end{aligned}$$

For completeness, let $z + T\mathbb{B}^d \subset K$ be any translated ellipsoid, where T is invertible. The two support inequalities for each symmetric slab in (3) imply

$$|\langle z, u_{jk}^{\varepsilon,\delta}\rangle| + \|T^* u_{jk}^{\varepsilon,\delta}\|_2 \le 1,$$

and hence $\|T^* u_{jk}^{\varepsilon,\delta}\|_2 \le 1$. Using the identity decomposition,

$$\operatorname{tr}(TT^*) = \sum_{j,k,\varepsilon,\delta} \alpha \|T^* u_{jk}^{\varepsilon,\delta}\|_2^2 \le \sum_{j,k,\varepsilon,\delta} \alpha = d.$$

AM–GM applied to the squared singular values of T yields

$$|\det T|^{2/d} \le \frac{\operatorname{tr}(TT^*)}{d} \le 1.$$

Thus $\mathrm{vol}_d(z + T\mathbb{B}^d) \le \mathrm{vol}_d(\mathbb{B}^d)$, proving that $\mathbb{B}^d$ is a largest-volume ellipsoid in K.

If $x = \sum_{j=1}^{s} x_j e_j \in F$, every inequality in (3) reduces to $p|x_j| \le 1$. Hence

$$K \cap F = \sqrt{\frac{d}{s}}[-1,1]^s,$$

and consequently

$$\frac{\mathrm{vol}_s(K \cap F)}{\mathrm{vol}_s(\mathbb{B}^d \cap F)} = \left(\frac{d}{s}\right)^{s/2} \frac{\mathrm{vol}_s([-1,1]^s)}{\mathrm{vol}_s(\mathbb{B}^s)}. \tag{4}$$

Finally write $d = \ell s + t$, where $0 < t < s$ because $s \nmid d$. Strict AM–GM applied to t copies of $\ell + 1$ and $s - t$ copies of ℓ gives

$$\left((\ell+1)^t \ell^{s-t}\right)^{1/s} < \frac{t(\ell+1) + (s-t)\ell}{s} = \frac{d}{s}.$$

Together with (2), this says $C_\square(d,s) < (d/s)^{s/2}$. Comparing with (4) proves the strict violation. $\square$

Smallest counterexample

Take $d = 3$, $s = 2$, $F = \operatorname{span}\{e_1, e_2\}$, $p = \sqrt{2/3}$, and $q = 1/\sqrt{3}$. Removing duplicate symmetric slabs, the body can be written

$$K = \left\{ x \in \mathbb{R}^3 : \begin{array}{ll} |px_1 + qx_3| \le 1, & |px_1 - qx_3| \le 1, \\ |px_2 + qx_3| \le 1, & |px_2 - qx_3| \le 1 \end{array} \right\}.$$

The proof above shows that $\mathbb{B}^3$ is its John ellipsoid, while

$$K \cap F = \sqrt{\frac{3}{2}}[-1, 1]^2, \qquad \operatorname{area}(K \cap F) = 6.$$

But $C_\square(3, 2) = \sqrt{2}$, so the conjectured unnormalized section bound is $4\sqrt{2} < 6$.

Verification notes

- The maximality proof includes arbitrary translations and arbitrary invertible T; it does not merely verify the necessary contact equations or local maximality.
- The construction gives a full John decomposition of $\mathrm{Id}_{\mathbb{R}^d}$, stronger than the projected decomposition required for the constrained fixed-axis problem.
- The supplied checker verifies unit norms, spanning, the weighted John identity, and the strict constant comparison for all $d \le 15$. It is a sanity check; the displayed identities are the proof.
- The conjecture remains sharp on the divisible pairs within this family: if $s \mid d$, then $C_\square(d, s) = (d/s)^{s/2}$.

Novelty check and limitations

A bounded search on 11 August 2026 checked the exact paper title, arXiv id, "Conjecture 5.10", "John ellipsoids of revolution counterexample", and close volume-ratio phrases on the arXiv domain. It recovered the source and the cube-section paper that supplies $C_\square$ [2], but no later resolution or counterexample. The local run indexes likewise contained no duplicate. This supports candidate novelty only; specialist author/citation review remains appropriate.

The theorem disproves the conjecture as written. It does not determine the sharp replacement constant for every centrally symmetric body, although the construction shows that no universal constant smaller than $(d/s)^{s/2}$ is possible. Since the source already proves the matching upper bound $(d/s)^{s/2}$, the same construction also identifies that bound as sharp for every $1 \le s < d$.

Human-review recommendation

Validity confidence is high. The main review points are: verify the four-sign outer-product calculation; check the support-function inequality for a translated ellipsoid; and confirm from the source definition that the unit ball is an admissible F-ellipsoid. Novelty confidence is moderate because the counterexample is elementary and should be checked with the source authors or a comprehensive citation search before publication.

References

[1] G. Ivanov, Z. Lángi, M. Naszódi, and À. Sagmeister, *John Ellipsoids of Revolution*, arXiv:2507.06947v2 (2025).

[2] G. Ivanov and I. Tsiutsiurupa, *On the volume of sections of the cube*, Analysis and Geometry in Metric Spaces **9** (2021), 1–18; arXiv:2004.02674.